

## Tutorial 9 – Recursion

1. Consider the following recursive function:

```
int func (int n) {
    if (n == 0 || n == 1)
        return 1;
    else
        return n * func( n - 1);
}
```

- Trace the program segment above and determine the result when  $n = 4$ . Use activation frames to show each of the steps taken.
  - What does the function *func* calculates?
  - What is the potential limitation of this recursive function *func*?
2. Use activation frames to evaluate the output of the following program.

```
#include <iostream>
using namespace std;

void sequence ( int number ) {
    if ( number < 13 ) {
        cout << number << endl;
        sequence( number + 4 );
    }
}

void main() {
    sequence(1);
    return 0;
}
```

3. With reference to the following main program, write a ***Fibonacci( )*** function and find its output when  $n$  is 3 using the activation frames based on the formula below:

$$F(n) = \begin{array}{ll} 0 & \text{if } n = 0; \\ 1 & \text{if } n = 1; \\ F(n-1) + F(n-2) & \text{if } n > 1; \end{array}$$

4. a) Prepare the activation frames for the following recursive function where the function call is **result = strange(4)**.

```
int strange (int n)
{
    int ans;
    if (n == 1)
        ans = 0;
    else
        ans = 1 + strange (n/2);
    return (ans);
}
```

- b) What is the value of **result** after the function call in (a) above?
5. Based on the following sample activation frame, write a C++ program using a recursion function **findMax** to find the max integer value in an array declared in the **main** function. Note that a max function is used in the function.

e.g.

```
n = 4
if(n==1) -> false
return max(arr[3], findMax(arr, 3))
```

:

:

:

```
n = 1
if(n=1) -> true
return arr[0]
```

Test the program using different number of integer values in an array. Ensure the maximum value is returned and displayed accordingly.

6. a) Write a recursive function **sumOfDigit(n)** to calculate and display the total of the digits in **n**.  
E.g. 1234 → 1 + 2 + 3 + 4 = 10.  
Hint: Use % and / operators in the function
- b) In the **main** function, declare a number with initial value 1234. The number of digits will be returned by the recursive function and displayed as a result.
- c) Draw the activation frames for the recursion function above and find its output.
7. a) Write a recursive function **countChar(string str, char ch, int index)** to count the number of occurrences of a character (ch) in a string (str). The index is referring to the value where the character is to be searched from in the string.  
E.g. In the string "programming", occurrences of m is 2
- b) In the **main** function, prompt the user to enter a string value and a character to count in the string. The number of digits will be returned by the recursive function and displayed as a result.